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United States Patent	12417312
Kind Code	B2
Date of Patent	September 16, 2025
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Constraint-based training data generation

Abstract

In one embodiment, a device may receive a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with metadata identifiers. The device may determine one or more training data constraints that restrict use of the application data as training data. The device may generate the training data in part by excluding application data of a particular type from being included in the training data based on a match between its metadata identifier and the one or more training data constraints. The device may provide the training data to be used to train a machine learning model.

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Appl. No.: 18/106600

Filed: February 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240265138 A1	Aug. 08, 2024

Publication Classification

Int. Cl.: G06F21/00 (20130101); G06F21/62 (20130101)

U.S. Cl.:

Field of Classification Search**CPC:** G06F (21/6254); G06F (21/629)**References Cited****U.S. PATENT DOCUMENTS**

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to constraint-based training data generation.

BACKGROUND

(2) Applications operate by handling data. For instance, executing an application can involve the storage, communication, processing, etc. of various types of data. The various types of data may include data whose handling is subject to various regulations. For example, data handling regulations at national, federal, state, industry, and/or organizational levels may be applicable to the data handled by an application. The enforcement of data compliance has only been made more complex and potential violations made more likely as applications are increasingly being developed as a set of distributed services running across a mix of multi-cloud and edge infrastructures and handling a mix of data types differentially subject to various regulations.

(3) For most organizations, the collection, use, and analysis of their data is crucial. For example, many organizations use data analysis and/or the use of production data to develop and train machine learning models that can be used to improve their operations and/or offer additional services or products.

(4) Unfortunately, collection and use of application data for model training largely occurs non-specifically and in a programmatic blind-spot with respect to enforcement of data compliance requirements. Therefore, given the current regulatory environment and trends, continuing to treat data compliance as an afterthought with respect to the collection and use of application data for model training will likely yield increased violations of data compliance regulations which may result in substantial fines, penalties, and/or other negative impacts to data handlers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments herein may be better understood by referring to the following description in conjunction with the accompanying drawings in which like reference numerals indicate identically or functionally similar elements, of which:

(2) FIGS. 1A-1B illustrate an example communication network;

(3) FIG. 2 illustrates an example network device/node;

(4) FIG. 3 illustrates an example architecture for data compliance;

(5) FIGS. 4A-4B illustrates an example architecture for data compliance and protection bindings;

(6) FIG. 5 illustrates an example architecture for constraint-based training data generation;

(7) FIGS. 6A-6B illustrate examples of model training deployments for constraint-based training data generation; and

(8) FIG. 7 illustrates an example simplified procedure for constraint-based training data generation.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

(9) According to one or more embodiments of the disclosure, a device may receive a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with metadata identifiers. The device may determine one or more training data constraints that restrict use of the application data as training data. The device may generate the training data in part by excluding application data of a particular type from being included in the training data based on a match between its metadata identifier and the one or more training data constraints. The device may provide the training data to be used to train a machine learning model.

Description

(10) A computer network is a geographically distributed collection of nodes interconnected by communication links and segments for transporting data between end nodes, such as personal computers and workstations, or other devices, such as sensors, etc. Many types of networks are available, with the types ranging from local area networks (LANs) to wide area networks (WANs). LANs typically connect the nodes over dedicated private communications links located in the same general physical location, such as a building or campus. WANs, on the other hand, typically connect geographically dispersed nodes over long-distance communications links, such as common carrier telephone lines, optical lightpaths, synchronous optical networks (SONET), or synchronous digital hierarchy (SDH) links, or Powerline Communications (PLC) such as IEEE 61334, IEEE P1901.2, and others. The Internet is an example of a WAN that connects disparate networks throughout the world, providing global communication between nodes on various networks. The nodes typically communicate over the network by exchanging discrete frames or packets of data according to predefined protocols, such as the Transmission Control Protocol/Internet Protocol (TCP/IP). In this context, a protocol consists of a set of rules defining how the nodes interact with each other. Computer networks may be further interconnected by an intermediate network node, such as a router, to extend the effective “size” of each network.

(11) Smart object networks, such as sensor networks, in particular, are a specific type of network having spatially distributed autonomous devices such as sensors, actuators, etc., that cooperatively monitor physical or environmental conditions at different locations, such as, e.g., energy/power consumption, resource consumption (e.g., water/gas/etc. for advanced metering infrastructure or “AMI” applications) temperature, pressure, vibration, sound, radiation, motion, pollutants, etc. Other types of smart objects include actuators, e.g., responsible for turning on/off an engine or perform any other actions. Sensor networks, a type of smart object network, are typically shared-media networks, such as wireless or PLC networks. That is, in addition to one or more sensors, each sensor device (node) in a sensor network may generally be equipped with a radio transceiver or other communication port such as PLC, a microcontroller, and an energy source, such as a battery. Often, smart object networks are considered field area networks (FANs), neighborhood area networks (NANs), personal area networks (PANs), etc. Generally, size and cost constraints on smart object nodes (e.g., sensors) result in corresponding constraints on resources such as energy, memory, computational speed and bandwidth.

(12) FIG. 1A is a schematic block diagram of an example computer network **100** illustratively comprising nodes/devices, such as a plurality of routers/devices interconnected by links or networks, as shown. For example, customer edge (CE) routers **110** may be interconnected with provider edge (PE) routers **120** (e.g., PE-1, PE-2, and PE-3) in order to communicate across a core network, such as an illustrative network backbone **130**. For example, routers **110**, **120** may be interconnected by the public Internet, a multiprotocol label switching (MPLS) virtual private network (VPN), or the like. Data packets **140** (e.g., traffic/messages) may be exchanged among the nodes/devices of the computer network **100** over links using predefined network communication protocols such as the Transmission Control Protocol/Internet Protocol (TCP/IP), User Datagram Protocol (UDP), Asynchronous Transfer Mode (ATM) protocol, Frame Relay protocol, or any other suitable protocol. Those skilled in the art will understand that any number of nodes, devices, links, etc. may be used in the computer network, and that the view shown herein is for simplicity.

(13) In some implementations, a router or a set of routers may be connected to a private network (e.g., dedicated leased lines, an optical network, etc.) or a virtual private network (VPN), such as an MPLS VPN thanks to a carrier network, via one or more links exhibiting very different network and service level agreement characteristics. For the sake of illustration, a given customer site may fall under any of the following categories:

(14) 1.) Site Type A: a site connected to the network (e.g., via a private or VPN link) using a single CE router and a single link, with potentially a backup link (e.g., a 3G/4G/5G/LTE backup

connection). For example, a particular CE router **110** shown in network **100** may support a given customer site, potentially also with a backup link, such as a wireless connection.

(15) 2.) Site Type B: a site connected to the network by the CE router via two primary links (e.g., from different Service Providers), with potentially a backup link (e.g., a 3G/4G/5G/LTE connection). A site of type B may itself be of different types:

(16) 2a.) Site Type B1: a site connected to the network using two MPLS VPN links (e.g., from different Service Providers), with potentially a backup link (e.g., a 3G/4G/5G/LTE connection).

(17) 2b.) Site Type B2: a site connected to the network using one MPLS VPN link and one link connected to the public Internet, with potentially a backup link (e.g., a 3G/4G/5G/LTE connection).

For example, a particular customer site may be connected to network **100** via PE-3 and via a separate Internet connection, potentially also with a wireless backup link.

(18) 2c.) Site Type B3: a site connected to the network using two links connected to the public Internet, with potentially a backup link (e.g., a 3G/4G/5G/LTE connection).

(19) Notably, MPLS VPN links are usually tied to a committed service level agreement, whereas Internet links may either have no service level agreement at all or a loose service level agreement (e.g., a “Gold Package” Internet service connection that guarantees a certain level of performance to a customer site).

(20) 3.) Site Type C: a site of type B (e.g., types B1, B2 or B3) but with more than one CE router (e.g., a first CE router connected to one link while a second CE router is connected to the other link), and potentially a backup link (e.g., a wireless 3G/4G/5G/LTE backup link). For example, a particular customer site may include a first CE router **110** connected to PE-2 and a second CE router **110** connected to PE-3.

(21) FIG. 1B illustrates an example of network **100** in greater detail, according to various embodiments. As shown, network backbone **130** may provide connectivity between devices located in different geographical areas and/or different types of local networks. For example, network **100** may comprise local/branch networks **160**, **162** that include devices/nodes **10-16** and devices/nodes **18-20**, respectively, as well as a data center/cloud environment **150** that includes servers **152-154**. Notably, local networks **160-162** and data center/cloud environment **150** may be located in different geographic locations.

(22) Servers **152-154** may include, in various embodiments, a network management server (NMS), a dynamic host configuration protocol (DHCP) server, a constrained application protocol (CoAP) server, an outage management system (OMS), an application policy infrastructure controller (APIC), an application server, etc. As would be appreciated, network **100** may include any number of local networks, data centers, cloud environments, devices/nodes, servers, etc.

(23) In some embodiments, the techniques herein may be applied to other network topologies and configurations. For example, the techniques herein may be applied to peering points with high-speed links, data centers, etc.

(24) According to various embodiments, a software-defined WAN (SD-WAN) may be used in network **100** to connect local network **160**, local network **162**, and data center/cloud environment **150**. In general, an SD-WAN uses a software defined networking (SDN)-based approach to instantiate tunnels on top of the physical network and control routing decisions, accordingly. For example, as noted above, one tunnel may connect router CE-2 at the edge of local network **160** to router CE-1 at the edge of data center/cloud environment **150** over an MPLS or Internet-based service provider network in backbone **130**. Similarly, a second tunnel may also connect these routers over a 4G/5G/LTE cellular service provider network. SD-WAN techniques allow the WAN functions to be virtualized, essentially forming a virtual connection between local network **160** and data center/cloud environment **150** on top of the various underlying connections. Another feature of SD-WAN is centralized management by a supervisory service that can monitor and adjust the various connections, as needed.

(25) FIG. 2 is a schematic block diagram of an example node/device **200** (e.g., an apparatus) that

may be used with one or more embodiments described herein, e.g., as any of the computing devices shown in FIGS. 1A-1B, particularly the PE routers **120**, CE routers **110**, nodes/device **10-20**, servers **152-154** (e.g., a network controller/supervisory service located in a data center, etc.), any other computing device that supports the operations of network **100** (e.g., switches, etc.), or any of the other devices referenced below. The device **200** may also be any other suitable type of device depending upon the type of network architecture in place, such as IoT nodes, etc. Device **200** comprises one or more network interfaces **210**, one or more processors **220**, and a memory **240** interconnected by a system bus **250**, and is powered by a power supply **260**.

(26) The network interfaces **210** include the mechanical, electrical, and signaling circuitry for communicating data over physical links coupled to the network **100**. The network interfaces may be configured to transmit and/or receive data using a variety of different communication protocols. Notably, a physical network interface **210** may also be used to implement one or more virtual network interfaces, such as for virtual private network (VPN) access, known to those skilled in the art.

(27) The memory **240** comprises a plurality of storage locations that are addressable by the processor(s) **220** and the network interfaces **210** for storing software programs and data structures associated with the embodiments described herein. The processor **220** may comprise necessary elements or logic adapted to execute the software programs and manipulate the data structures **245**. An operating system **242** (e.g., the Internetworking Operating System, or IOS®, of Cisco Systems, Inc., another operating system, etc.), portions of which are typically resident in memory **240** and executed by the processor(s), functionally organizes the node by, inter alia, invoking network operations in support of software processors and/or services executing on the device. These software processors and/or services may comprise training data generation process **248**, as described herein, any of which may alternatively be located within individual network interfaces.

(28) FIG. 3 illustrates an example architecture **300** for data compliance, according to various embodiments. The architecture **300** may include a data compliance process **328**. Data compliance process **328** may be utilized to provide configuration, observability, and enforcement of data compliance rules. Data compliance process **328** may accomplish these functions utilizing Data Compliance as Code (DCaC).

(29) DCaC may include integrating a data compliance mechanism into the program code of the application. For example, data compliance process **328** may be utilized to build data compliance into the application development process, supported by automated code annotations, bindings between such annotations and categories of sensitive data, and controls at code, build, and pre-deploy time. Data compliance process **328** may provide a mechanism whereby application developers proactively assist data teams, application managers, and legal departments with data compliance, while ensuring that developers may remain oblivious to specific regulations, data related obligations, or compliance requirements that organizations might have across different regions.

(30) For example, data compliance process **328** may include data annotating process **302**. Data annotating process **302** may facilitate application developer **312** automatically adding metadata to program code of an application **316** during the development of the application **316**. In various embodiments, this may be performed by automated annotations of data fields in the program code and by the creation of references to such annotations at code-build time. These references to annotated code may be automatically rendered in the form of machine-readable data manifest **314**.

(31) More specifically, data annotating process **302** may provide a mechanism for automated annotations of the program code of application **316**, including classes, application programming interfaces (APIs), and the resulting data at code/build time (e.g., by implementing a Low-Code/No-Code approach supported by software development kits (SDKs) **320** and tooling **318**). Application developers may utilize SDKs **320** and tooling **318** to automatically label data topics, data producers, data consumers, data processors, data holders, etc. For instance, developers may label

certain data by annotating it with a data type identifier. For example, a developer may annotate certain data as “protected-type-1,” or other data as “protected-type-2,” and so on.

(32) SDKs **320** in data annotating process **302** may provide a set of predefined data types out-of-the-box, including associations by default to specific categories of sensitive data. Sensitive data may include a type of data that may be considered personal, private, sensitive, confidential, protected, secret, restricted, personally identifiable information (PII), etc. In some examples, sensitive data may include data that is subject to regulation. For example, Table 1 lists examples of predefined protected data types and default associations to some examples of categories of sensitive data.

(33) TABLE-US-00001 TABLE 1 PROTECTED DATA TYPE DEFAULT ASSOCIATION
protected-type-1 Customer PII protected-type-2 Employee PII protected-type-23 Patient Analysis Results protected-type-41 Sales Confidential protected-type-56 Restricted HR unprotected NA

A list of the associations, such as the example illustrated in Table 1, may provide associations by default to several categories of sensitive data, including but not limited to PII, confidential, restricted, and unprotected data. In some embodiments, the set of predefined protected data types might be standardized or rely on an existing taxonomy.

(34) SDKs **320** in data annotating process **302** may also provide a mechanism to define and use custom data types in annotating program data of the application **316**. For example, custom data types may be utilized, which identify protected data types that are not covered by any of those available by default in SDKs **320**. For example, “custom-type-1” might be a custom data type associated to a category of sensitive data such as “Restricted Employee Poll.” In various embodiments, the generation and/or insertion of the annotations into the program code of the application **316** may be accomplished by an automated process (e.g., a programmatic identification of data of a particular data type triggering an automated insertion of an annotation of the data as the particular data type, etc.), a partially automated process (e.g., a programmatic flagging of data of a particular data type with a supervised or manual annotation of the data as the particular data type, etc.), and/or a manual process (e.g., a manual flagging of data of a particular data type and/or a manual annotation of the data as the particular data type, etc.).

(35) In various embodiments, associations between protected data types and categories of sensitive data may be assigned and/or instrumented by different organizations and at different moments in time. In some cases, the association between protected data types and categories of sensitive data may be assigned by application developers **312** at code/build time. This might be the case when the team of application developers **312** is part of, or develops for, the organization that may use or manage the application **316**. In such cases, the team of application developers **312** might have sufficient knowledge about the data and their use, so that they may either use the associations provided by default or create custom ones.

(36) In additional instances, application developers **312** of application **316** and/or the users of the application **316** might belong to different organizations. For example, this may be the case when application developers **312** are a DevSecOps team that develops an application **316** that may be used across different organizations, industries, etc. In such cases, application developers **312** may be unaware of the categories of data that should be assigned by a data team and/or application manager **304** in another organization (e.g., precisely what data is confidential and what data is not with respect to that organization and its use of the application **316**). In these instances, application developers **312** may leverage SDKs **320** and tooling **318** to approach data labeling and association in a manner that sidesteps the knowledge deficit while still instilling the functionality. For example, the application developers **312** may leverage SDKs **320** and tooling **318** to automatically add the different classes of protected data type at code and build time (e.g., utilizing predefined and custom protected data types). Additionally, or alternatively, the application developers **312** may leverage SDKs **320** and tooling **318** to automatically insert references in the form of machine-readable

descriptions for the protected data types that may be used to generate data manifest **314** bound to application **316** at build time.

(37) The protected data type annotations and their corresponding references may be utilized by a data team and/or application manager **304** in another organization to select and/or create automated associations **326** between categories of annotated data in the application **316** (e.g., metadata provided by application developers **312**) and specific categories of sensitive data (e.g., personal data, private data, sensitive data, confidential data, protected data, secret data, restricted data, PII, etc.). For instance, each protected data type might be bonded to a class of tokens (e.g., JSON Web Tokens with a specific scope), which in turn might represent different categories of sensitive data for a data team and/or application manager **304**.

(38) In a specific example, an API call for application **316** may be labeled by application developers **312** with a data type identifier such as “custom-type-7” at code/build time. The “custom-type-7” labeled API call may attempt to access certain data using its bound token (e.g., “Token 7”) with a scope defined by, for example, a data team and/or application manager **304** before application **316** was deployed. From the data team and/or application manager **304** perspective, the attempt to access this data may translate to a request to access, for instance, “Confidential Partner” data. As such, the data type labels, and their associations may be utilized as an automated data mapping between the programmatic operations of application **316** and the sensitive data implicated in those operations. In various embodiments, these associations and functionalities may be supported by compliance engine **306** based on the selection, configuration, and automation of data compliance rules before application **316** is deployed and/or post-deployment.

(39) In some examples, application developers **312**, which again may be a DevSecOps team, might opt for a hybrid approach to generating these associations. For example, this may be the case when making some custom associations between data types and categories of sensitive data or using those predefined in the system (e.g., “protected-type-1” to “Customer PII”) might not only be trivial for the application developers **312** but also may facilitate the task of a data team and/or application manager **304** in defining associations. However, other associations might not be apparent to application developers **312**. Hence, certain data in application **316** may be labeled as “protected types” along with their corresponding machine-readable descriptions in data manifest **314**, though they may remain unassigned to a specific category of sensitive data, so they can be associated later by a data team and/or application manager **304** before the application is deployed, or by an automated data lineage, classification, and tagging process at run time (e.g., during the testing phase, that is, before the application is deployed in production).

(40) In some embodiments, a data team and/or application manager **304** may be provided with a mechanism to change the associations created by application developers **312** or even associate more than one category of sensitive data to a given data type (e.g., a data team and/or application manager **304** may associate certain data with both “Employee PII” and “Confidential Data”). Hence, two categories of data compliance policies (e.g., one for “Employee PII” and another for “Confidential Data”) may apply and restrict even further the access to this category of data. In general, a data team and/or application manager **304** may be able to Create, Read, Update, or Delete (CRUD) any association between the metadata provided by application developers **312** and categories of sensitive data.

(41) In various embodiments, a data team and/or application manager **304** may proactively create a set of custom data types. A data team and/or application manager **304** may provide the set of custom data types to application developers **312**. Application developers **312** may then utilize the set of custom data types so that application **316** is annotated at development based on guidelines (e.g., the set of custom data types, etc.) provided beforehand by the data team and/or application manager **304**.

(42) In additional embodiments, application developers **312** and a data team and/or application

manager **304** may collaborate to annotate application **316**. For example, application developers **312** and a data team and/or application manager **304** may iterate in the annotation and association processing in an agile manner. For example, the iteration may be performed as part of a continuous integration/continuous delivery (CI/CD) pipeline (e.g., at testing, staging, and production).

(43) In some examples, application **316** may be composed of several services developed with different programming languages. Therefore, application **316** may utilize different SDKs **320**. In some instances, the annotation methods and terminology applied to application **316** may vary depending on the programming language (e.g., usually referred to as attributes in C#, decorators in Python, annotations in Golang, etc.). In such cases, tooling **318** of data annotating process **302** may examine the different predefined and custom data types used with different SDKs **320**, perform checks, and ensure consistency in the annotations and enumeration across the different services at build time. For example, these consistency checks may ensure that a given “custom-type-X” data type identifier represents the same type of data across services programmed using different programming languages even if they were programmed by different developers. Overall, the data annotating process **302** may provide different degrees of freedom to application developers **312**, data teams and/or application managers **304**, and the number of protected data types used, and their corresponding associations may vary depending on the type of application **316**.

(44) Data annotating process **302** may, as described above, be utilized in generating automated data references. Specifically, data annotating process **302** may automatically render a data manifest **314** bonded to application **316** at build time. Data manifest **314** may provide machine-readable descriptions of the predefined and/or custom data types used by application **316**. A combination of SDKs **320** and tooling **318** may facilitate the instrumentation and automation of the program code at build time, including the automated rendering of data manifest **314**. In some cases, application **316** may be composed of various containers. Each container may be built and packaged with its own data manifest, such that the final data manifest rendered for application **316** may be a composition of the individual data manifests. In some cases, application **316** may include dependencies on external services, such as a MySQL database. Such dependencies may be captured as a dependency manifest. Data fed, processed, retained, or retrieved from these external services may also be annotated and automatically captured in application **316** data manifest **314**.

(45) Data annotating process **302** may, as described above, be utilized for decoupling data compliance from the business logic of application **316**. For example, SDKs **320** and tooling **318** of data annotating process **302** may provide automated mechanisms for decoupling the configuration, observability, and enforcement of data compliance rules from the business logic of application **316**. In some instances, application **316** may be a cloud/edge native application, which may be implemented as a set of workloads composing a service mesh. The decoupling of data compliance from the business logic may be especially relevant for applications of this type, as geographically dispersed and/or variably deployed workloads may implicate increased data compliance complexity.

(46) Various possible embodiments for decoupling data compliance from the business logic of application **316** may be utilized. For instance, a sidecar model, where the services that implement the business logic of application **316** are deployed together with sidecar proxies associated to each of those services, may be utilized. The sidecar proxies may be utilized to enforce horizontal functions that are independent of the business logic, such as routing, security, load balancing, telemetry, retries, etc. As such, the sidecars may be well-positioned to decouple, observe, and control data compliance. For example, a combination of distributed data compliance controllers and sidecar proxies may be used to configure, observe, and enforce data compliance rules across different geographies, and distributed multi-cloud and edge infrastructures **334**.

(47) Instead of, or in addition to, using sidecars, various embodiments may use client libraries, daemons working in tandem with the application-specific services, or sandboxed programs in the OS kernel, e.g., using the Extended Berkeley Packet Filter (eBPF). Further embodiments may use

an agentless approach or embed such functionality in a container orchestration system, such as Kubernetes itself. In any case, the functionality introduced herein may enable the portability and reuse of observability and enforcement of data compliance functions across not only different applications but also cloud and edge environments.

(48) The above-described data annotating process **302** may yield a portable annotated application **316** that is geared with built-in annotations for different types of protected data. In addition, the yielded annotated application **316** may be structured to operate while remaining agnostic of any state, country, industry, organization-specific regulation and/or data policy requirements that a data team and/or application manager **304** might have. As a result, data annotating process **302** may be leveraged as a new model of building applications including DCaC by not only data teams and/or application managers **304**, but also software as a service (SaaS) providers and others.

(49) Data compliance process **328** may provide configuration, observability, and enforcement of data compliance rules. As described above, associations **326** between categories of annotated data in application **316** and specific categories of sensitive data may be instrumented prior to a deployment of application **316**. The associations **326** may be used to control the processing and use of data during and after the deployment of application **316**. More specifically, compliance engine **306** may utilize associations **326** together with current data compliance regulations governing data handling in each region where application **316** may be used, as well as a specific organization's compliance rules **308** for/while using application **316**, to enforce compliance with them. Such controls may apply to data access requests, data storage and retention policies, data processing requirements, etc. of application **316** both at deploy and execution time, etc.

(50) To this end, data compliance process **328** may include data compliance regulation repository **322**. Data compliance regulation repository **322** may provide a repository of data compliance rules. For example, data compliance regulation repository **322** may include a repository of industry regulations **324** which may be applicable to the use of application **316**. For example, with respect to instances where application **316** is used by a healthcare provider, data compliance regulation repository **322** may include industry regulations **324** such as Health Insurance Portability and Accountability Act of 1996 (HIPAA) regulations applicable to handling of data in the healthcare industry. In other examples, data compliance regulation repository **322** may include a repository of national regulations **330** which may be applicable to the use of application **316**. For example, with respect to instances where application **316** is based in a member state of the E.U., data compliance regulation repository **322** may include national regulations **330** such as the GDPR applicable to handling of data in the E.U.

(51) The data compliance regulations included in data compliance regulation repository **322** may be consumed by a data team and/or application manager **304** as a service (aaS). Data compliance regulation repository **322** may support input, expression, collection, approval, visualization, and/or use of data compliance policies covering multiple categories of rules. For example, data compliance regulation repository **322** may store data compliance policies that are specific to an industry, those that may apply at a national, multi-national, federal, state, and industry levels, etc. For instance, an organization (e.g., a multi-national company) may leverage a data compliance regulation repository **322** service of a data compliance process **328** and utilize the regulations already available in data compliance regulation repository **322**, which may cover regulations across several industries and countries out-of-the-box. An organization may select the target state, country or region, the industry if needed, and select the data compliance regulations that may be applicable at the organizational level (e.g., organization's compliance rules **308**).

(52) Compliance engine **306** may offer APIs and a user-friendly user interface (UI) through which a data team and/or application manager **304** may select and define data compliance requirements. For instance, if application **316**, which handles Customer PII data, needs to be deployed in British Columbia, Canada, a data team and/or application manager **304** may simply select “Customer PII.fwdarw.Apply Local Regulation” to constrain the processing, storage, retention, and access to

Customer PII data according to the regulations in British Columbia as retrieved from data compliance regulation repository **322**. To this end, compliance engine **306** may compute and handle the resulting constraints that apply to Customer PII data in British Columbia transparently to data teams and/or application managers **304**. More specifically, the set of data compliance constraints may be captured in a machine-readable format from data compliance regulation repository **322**, and therefore, used by compliance engine **306** programmatically.

(53) In some examples, compliance engine **306** may be utilized as a pluggable module working in tandem with one or more workload engines **332**, such as Cisco Intersight or any automation tool offered by a hyperscaler, or other cloud and edge providers. Workload engines **332** may manage the deployment of application **316**, subject to the rules and constraints provided by compliance engine **306**.

(54) In various embodiments, compliance engine **306** may operate either in a push or a pull model. For instance, in a pull model, a workload engine **332** may receive a request to deploy application **316** in a given region (e.g., a request from a site reliability engineering (SRE) and/or information technology (IT) team **310**). In such a case, workload engine **332** may issue a request to compliance engine **306**, to compute and return data compliance rules and constraints that must be applied for their specific deployment. Alternatively, in a push model, a data team and/or application manager **304** may select the compliance rules required and a declarative intent for application deployment may be issued from compliance engine **306** to one or more workload engines **332**. Such deployments may involve multi-cluster service meshes, which may run across nodes hosted in various infrastructures, including multi-cloud and edge infrastructures **334**.

(55) In various embodiments, the nodes hosted in the various infrastructures (e.g., edge infrastructures **334**) may also operate as data collection nodes. For example, an application may be executed at and/or in association with the node and application data associated with that execution may be collected at the data collection node. The collection, sampling, modification, and use of third-party application data for model training purposes may be supported by one or more dataset generator processes at and/or associated with the data collection node.

(56) FIGS. **4A-4B** illustrate an example architecture **400** for data compliance and protection bindings according to various embodiments. Architecture **400** may be utilized to deliver DCaC. For example, architecture **400** may include compliance engine **306**. Compliance engine **306** may include data compliance rules module **402**, compliance intent engine **404**, and/or observability and assurance engine **406**.

(57) Data compliance rules module **402** may compute compliance constraints based on a combination of inputs. For example, the constraints may be computed based on a combination of inputs including the target state, country, or multi-country region for an application, the industry the application is being utilized within, and/or the compliance rules required by an organization using the application.

(58) Compliance intent engine **404** may include an association service **408**. Association service **408** may manage a set of association tables **410**. Association service **408** may include functionality to allow a data team and/or application manager **304** to select, configure, and create the associations (e.g., associations **326** from FIG. **3**) and store them in the form of protection bindings **424** in association tables **410** (e.g., populated association table **422** and its associated protection bindings **424** provide a non-limiting specific example of one such association table). The protection bindings **424** may define a data handling scope bonded to the association between a data type and its associated category of sensitive data. The data handling scope may be an indication of how data will be handled by the application (e.g., applicable tokens, token scopes, encryption keys, etc.). The protection bindings **424**, stored in populated association table **422** may be created and/or maintained by compliance engine **306**. The protection bindings **424** may not be reinserted into the program code but rather maintained by compliance engine **306** since protection bindings **424** may be subject to changes over time (e.g., the scopes might change, encryption keys may be rotated,

etc.) and keeping the protection bindings **424** outside of the program code may prevent these changes from affecting the program code.

(59) Additionally, compliance intent engine **404** may include a service intent engine **412**. Service intent engine **412** may interface with workload engines **332**. Service intent engine **412** may either receive and/or process data compliance requests from workload engines **332** (e.g., pull model) or create and/or issue declarative intents encoding a deployment request to workload engines **332** (e.g., push model).

(60) Observability and assurance engine **406** may receive telemetry data from services deployed in the field (e.g., from a service mesh deployed across multi-cloud and edge infrastructures **334**). In addition, observability and assurance engine **406** may push data compliance configurations and data traffic filters in real-time **336** out to workloads deployed in the multi-cloud and edge infrastructures **334**.

(61) To populate association tables **410** and create protection bindings **424**, association service **408** may obtain inputs defining the associations of the protected data types and/or their data handling scopes. For example, for each annotated application **316**, association service **408** may obtain data manifests **314** describing the set of predefined and custom protected data types handled by annotated applications **316** and potential associations already made by application developers (e.g., application developer **312** in FIG. 3).

(62) Additionally, association service **408** may obtain data handling scopes to be bonded to individual associations between given data types and their associated category of sensitive data. For example, association service **408** may obtain, as an input, categories of tokens and corresponding scopes that may be associated with each of the protected data types used in the annotated classes and methods that compose the annotated application **316**. Such tokens may be defined and/or obtained from external token management service **416** (e.g., from OKTA).

(63) Further, association service **408** may obtain, as an input, identifiers of encryption keys. The encryption keys may be those keys associated with each of the protected data types used in the annotated classes and methods that compose annotated applications **316**. For example, different categories of sensitive data may be encrypted utilizing techniques such as bring your own key (BYOK) or hold your own key (HYOK). The encryption key identifiers (ID) may be defined or obtained from external encryption key service **418**, which may extract encryption key IDs. External encryption key service **418** may interface with key management service (KMS) **420** and may create references (e.g., key IDs) to encryption keys stored and managed by KMS **420**. In this manner, the keys may not be managed by association service **408**, but instead may remain secure with KMS **420**.

(64) This set of inputs may be utilized by association service **408** to populate association tables **410**. In some examples, there may be one association table populated per annotated application **316** and/or per data compliance zone (e.g., a geographical area where the application is deployed, etc.). A populated association table **422** may include the automatically associated annotated data types (e.g., protected data type labels), with categories of sensitive data (e.g., encoded in the form of tokens with specific scopes as illustrated in populated association table **422**), along with pointers to the encryption keys used for each category of protected data (e.g., key IDs).

(65) Compliance engine **306** may handle one populated association table **422** per compliance zone for each annotated application **316**. In some embodiments, populated association table **422** may be extended to include columns denoting API paths to access the data resources. In some cases, the paths may explicitly embed the protected data types used by the application developers. In addition, the tokens, scopes, and key IDs may be renewed and dynamically updated in populated association table **422** depending on their validity and/or expiration time. Populated association table **422** may also manage more advanced associations, including men bindings (e.g., two or more protected data types might be associated to a single token/scope class).

(66) Once association tables **410** are defined and/or populated, a data team and/or application

manager **304** may select the organization's compliance rules **308**. Organization's compliance rules **308** may include data compliance rules (e.g., data compliance rules **414**) selected to be required for a specific annotated application **316** (e.g., "App 1") that should be deployed in a specific geographical region (e.g., "Compliance Zone 1").

(67) For example, data compliance rules **414** may include constraints selected to be applied to each category of sensitive data (e.g., "Researcher PII," "Patient Analysis Results," "Study-Confidential Class 1." etc.) within annotated application **316** "App 1". Data compliance rules **414** may further specify the selected constraints associated with processing the category of sensitive data, storage, and retention of the category of sensitive data, and/or accessing or viewing of the category of sensitive data by the annotated application **316**.

(68) Data compliance rules **414** may be rules that may apply to a new annotated application **316** (e.g., "App 1"). In this example, application "App 1" may handle data about clinical trials of a new drug developed by a pharmaceutical company. Application "App 1" may collect and analyze data and provide insights about the new drug. In this example, the application "App 1" may support several categories of sensitive data.

(69) For example, the application "App 1" may support the category of sensitive data "Researcher PII" data, which may include PII of the employee conducting one of the trials. In this example, the rule chosen may be one to restrict the processing, storage, retention, and access to the data according to a "local regulation." For example, if the compliance zone where the application is going to be deployed is British Columbia, Canada, then this rule automatically constrains the location of workloads, the storage, and any attempt to access or view such data according to the regulation on PII in British Columbia, Canada. If the trials were instead to be conducted in New Delhi, India, the rule might be the same (e.g., applying a local regulation), but the resulting constraints yielded by compliance engine **306** will usually differ from those in British Columbia, Canada by virtue of the two locations having different data handling regulations.

(70) Application "App 1" may also support the category of sensitive data "Patient Analysis Results" data. In this example, the rule chosen by a data team and/or application manager **304** may adhere to both the local and industry-specific regulation. In the examples of application "App 1." the industry-specific regulation may include, for example, specific legislation constraining the processing, storage, retention, and access to patients' data with respect to clinical trials.

(71) Application "App 1" may additionally support the category of sensitive data "Study-Confidential Class 1" data. This data type may be a custom protected data type that may be implemented to enable researchers to keep a specific category of data related to the clinical trials as highly confidential. In this example, the processing, retention, and access to the data may be constrained to a specific facility. For example, perhaps the processing, storage, retention, and access of the "Study-Confidential Class 1" data is constrained to the premises of a "Laboratory 1" associated with the clinical trial (e.g., "On prem-L1").

(72) Based on data compliance rules **414** input along with the corresponding protection bindings **424**, data compliance rules module **402** may identify the implicated categories of sensitive data. For example, the three categories of sensitive data (e.g., "Employee PII," "Patient Analysis Results," and "Study-Confidential Class 1") listed in data compliance rules **414**, along with corresponding protection bindings **424** in populated association table **422** may be identified by data compliance rules module **402**. These categories of sensitive data may be defined and/or used by a data team and/or application manager **304** and/or may have already been associated to specific predefined and custom protected data types, such as the ones shown in populated association table **422** and/or protection binding **424**.

(73) In addition, data compliance rules module **402** may identify the selected compliance requirements listed in data compliance rules **414**. For example, data compliance rules module **402** may identify the compliance requirements specified for processing, storage, retention, and access for each of the categories of sensitive data as defined in data compliance rules **414**.

- (74) Data compliance rules module **402** may compute the set of compliance constraints that apply to application “App 1” based on data compliance rules **414** and/or a compliance zone selected (e.g., a target country and industry for “App 1”). In some examples, the set of compliance constraints may be computed from, for example, industry regulations **324**, national regulations **330**, etc. obtained from data compliance regulation repository **322**.
- (75) The output of data compliance rules module **402** (e.g., the computed set of compliance constraints for a category of sensitive data) may be processed by compliance intent engine **404**. Compliance intent engine **404** may link the resulting constraints to the corresponding populated association table **422** and send this output to both observability and assurance engine **406** and service intent engine **412**. As such, the compliance constraints may be linked to categories of sensitive data and/or their associated protected data types in the program code. Therefore, the compliance constraints may be linked to individual portions of the application code. For instance, the constraints may be linked to control a data transfer through an API call that was previously annotated by application developers **312** using the protected data types referenced in protection bindings **424**, populated in association table **422**, and constrained by data compliance rule module **402** according to data compliance rules **414**.
- (76) In various embodiments, once a service mesh is deployed, a data consumer process may request access to a data resource through an API. This may be implemented using a GET method including a path containing the field “custom-type-1” (CT1), which, according to populated association table **422**, represents the custom protected type “Study-Confidential Class 1.” The HTTP request may be transported and forwarded over mTLS across the sidecar proxies in the service mesh. The authorization header in the service mesh may carry “token 3” with a specific scope “scope 3,” as defined in populated association table **422**. In this example, “scope 3” represents the category of sensitive data “Study-Confidential Class 1,” and the constraint in this case is that the data of that type must be retained on “prem Lab1.” To that end, service intent engine **412** may have requested and/or instructed a workload engine **332** to deploy the workloads handling “CT1” “on prem L1.” In turn, observability and assurance engine **406** may have configured data filters in the sidecar proxies to enforce access control. For instance, API calls using an authorization token with “scope 3” may be restricted to data consumers located “on prem L1.”
- (77) Compliance engine **306** may process more elaborate data compliance rules than simply those illustrated, including the selection of specific locations for processing, storage, retention, and access for each category of sensitive data. For instance, a data team and/or application manager **304** may choose a specific data center (e.g., where their data warehouse is hosted), a compliant public or edge zone, a compliant private cloud or edge site, combinations of these, etc. While some of these selections may be very specific, others might remain openly declarative, which a member of SRE/IT team **310** may translate into a specific infrastructure request for deploying the application, or some of the services that comprise the application.
- (78) Some of the tokenization mechanisms described herein may be externally handled, such as by an authorization server, which may potentially work in concert with a delegated authorization solution (e.g., OAuth 2.0/OpenID Connect), a single sign-on (SSO) solution, etc. In such scenarios, the specific categories of tokens and scopes references in association tables **410** may be obtained from external systems.
- (79) In addition to the tokenization mechanisms, a data team and/or application manager **304** may also select the user and/or process groups that may have access to the different categories of sensitive data (e.g., read only, read and write, or no access). Such groups may be managed using internal tools or they may be externally handled by an authorization service (e.g., OKTA). In some embodiments, the selection of token scopes and the access rights applied to user and/or process groups may be made jointly. These additional constraints may also be part of the data compliance rules and state maintained by data compliance rule module **402**.
- (80) In an alternative embodiment, compliance engine **306** may also enable the use of third-party

annotations and/or data catalogs (e.g., imported from external data classification and tagging systems, such as from Collibra, OneTrust, or others). In such cases, the protected types used in protection bindings **424**, and populated in association table **422**, may be comprised of a set of annotation labels (i.e., metadata) added by application developers **312** (e.g., PT2, PT23 and CT1 in FIGS. **4A-4B**) as well as third-party labels provided by external systems.

(81) As such, DCaC implemented through architecture **400** may provide two levels of decoupling. First, a decoupling between the annotations or metadata embedded in the program code of the application as provided by application developers **312** at code/build time and the categories of sensitive data that may be selected and associated by a data team and/or application manager **304** before the annotated application **316** is deployed. Second, a decoupling between the categories of sensitive data and the rules selected by a data team and/or application manager **304**, and the specific data compliance regulation and data compliance constraints that may apply to a given industry and/or region.

(82) Such an approach may facilitate application developers **312** proactively assisting a data team and/or application manager **304**, while all of them are allowed to remain oblivious to the specificities and intricacies of the different data compliance regulations across the different industries and regions.

(83) It will be apparent to those skilled in the art that other processor and memory types, including various computer-readable media, may be used to store and execute program instructions pertaining to the techniques described herein. Also, while the description illustrates various processes, it is expressly contemplated that various processes may be embodied as modules configured to operate in accordance with the techniques herein (e.g., according to the functionality of a similar process). Further, while processes may be shown and/or described separately, those skilled in the art will appreciate that processes may be routines or modules within other processes.

(84) As noted above, the number of laws, regulations, and rules regarding the storage and use of certain types of data are continually increasing across the globe. For instance, the General Data Protection Regulation (GDPR) in Europe places strict regulations on how a user's personal data is collected and shared. These and other regulations have spawned independent efforts across several countries to ensure that online applications comply with specific data regulations at national, federal, or state level, and particularly, those that are cloud-based. This acceleration in data sovereignty regulations is posing complex challenges to the organizations that use or manage that data, since legal obligations and constraints vary from country to country. The challenge is even greater since data compliance requirements are often not limited to data sovereignty obligations. For example, depending on the type of an application, data compliance may demand the amalgamation of other regulations, such as industry-specific regulation (e.g., complying with HIPAA obligations in the healthcare industry in the United States), or organization-specific rules (e.g., on how to deal with confidential data). Regardless, of the source of the data compliance requirement, it's observance may involve the handling of data in a particular way at runtime.

(85) For most organizations, the collection, use, and analysis of data is crucial. Hence, one of the areas where these challenges are evident is data analysis and the use of production data for model training. Previous approaches have developed numerous data analysis techniques, many of which require datasets to train their models, learn, and improve the inference algorithms (e.g., using supervised, semi-supervised or unsupervised learning techniques). Indeed, many organizations already have valuable datasets in production, which they are willing to use for model training purposes.

(86) Commonly, the data used for training the model are collected and, in many cases, sampled, which are then sent to the cloud for centralized learning (e.g., using centralized models running in the cloud). In this scenario, the raw data is sent to the cloud for training purposes, thereby leveraging the compute power and scalability offered by cloud infrastructures. However, basic data compliance aspects remain a challenge. For instance: Is model training compliant with the primary

purpose of data collection and utilization according to regulatory obligations? Is the data transfer to the cloud and subsequent processing compliant with data regulations? Is it compliant with the organization's rules?

(87) Some approaches rely on a federated learning configuration. In this configuration, the data used for the training may stay where it is collected and possibly sampled, which may then be used by federated learning agents to train local models (e.g., at the edge). These local models may be combined into a global model, which may also leverage the benefits offered by cloud infrastructures. However, this configuration has two main problems; it is subject to bias and accuracy challenges (including dataset balancing issues), and, as in the case of the centralized configuration, it suffers from potential compliancy issues associated to the purpose and use of the data (e.g., secondary purpose issues).

(88) Therefore, data compliance is presently a missing piece in the field of model training. Indeed, data compliance has been an “afterthought”, as data compliance is only being considered once the datasets need to be transferred and/or used for model training. For example, existing data compliance strategies lack integration in the application program code itself (e.g., they do not utilize a DCaC model) and instead rely on dynamically trying to determine what type of data is being shared and/or used.

(89) As such, existing strategies for generating datasets to train models, learn, and improve inference algorithms provide insufficient data visibility and/or control at collection nodes to navigate the modern minefield of data compliance regulations. Accordingly, these strategies are increasingly yielding data compliance violations resulting in substantial fines, penalties, and/or other negative impacts to data handlers.

Constraint-Based Training Data Generation

(90) The techniques herein introduce mechanisms for the collection and sampling of production (i.e., real) data handled by a DCaC-enabled application and automate the generation of compliant datasets that can be transferred to the cloud for centralized training and learning (or to other locations). This capability may be based on observing and sampling production data and their associated metadata identifiers (e.g., tags created and associated to different types of sensitive data as part of the DCaC process), along with a compliance-led dataset generation technique, which may selectively combine synthetic, anonymized, and original data.

(91) Illustratively, the techniques described herein may be performed by hardware, software, and/or firmware, such as in accordance with training data generation process **248**, which may include computer executable instructions executed by the processor **220** (or independent processor of interfaces **210**) to cause the performance of functions relating to the techniques described herein.

(92) Specifically, according to various embodiments, a device may receive a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with metadata identifiers. The device may determine one or more training data constraints that restrict use of the application data as training data. The device may generate the training data in part by excluding application data of a particular type from being included in the training data based on a match between its metadata identifier and the one or more training data constraints. The device may provide the training data to be used to train a machine learning model.

(93) Operationally, FIG. 5 illustrates an example architecture **500** for constraint-based training data generation, according to various embodiments. As previously mentioned, a workload engine may manage deployment of an application **512**, subject to the rules and constraints provided by compliance engine **306**. Again, such deployments may involve multi-cluster service meshes, which may run across nodes hosted in various infrastructures, including edge infrastructures (e.g., edge infrastructures **334** of FIG. 3).

(94) Therefore, the nodes where the application **512** is executing may operate as data collection nodes **506** where application data associated with application **512** is collected from one or more

data sources **510**. As detailed below, the data collection nodes **506** may also run another application, such as a compliant dataset generator **508**. In some embodiments, compliant dataset generator **508** may include a process executed (e.g., entirely, partially, etc.) locally to and/or by the data collection node **506**. Alternatively, the process may be executed (e.g., entirely, partially, etc.) remotely from data collection node **506** such as at a different node and/or node managing device. (95) Compliant dataset generator **508** may support the collection, sampling, modification, and use of third-party application data for model training purposes, in a compliant way. To this end, compliant dataset generator **508** may utilize an interface **530** with compliance engine **306**, in order to determine the data compliance constraints that may apply depending on the categories of data collected, the region and industry, and the compliance rules defined by an organization's compliance rules **308**. Once a compliant dataset is generated, the dataset may be sent to a centralized and compliant learning framework running in a cloud **528**.

(96) For example, compliant dataset generator **508** may run in a data collection node **506** (e.g., in a computing node at the network edge). A data training team **504** may manage the configuration of compliant dataset generator **508**. The data training team **504** and the application managers **304** may be affiliated or not to the same organization, and they may coordinate but operate independently.

(97) At operation **542**, a configuration/request for training data may be received by compliant dataset generator **508**. The request may be received from data training team **504**. The configuration/request may specify a configuration to be applied to compliant dataset generator **508** using configurator **516**. The configuration may stipulate an identity of the application **512** that may provide the data to train the models. For example, the stipulated identity may be the identity of the application **512** running in the data collection node **506** that may handle data in production which may be required for centralized model training.

(98) Application **512** and its corresponding data may have been annotated using DCaC, and therefore, the data manifest bonded to application **512** may be available to compliance engine **306** as outlined above. Application **512** and/or its corresponding data may be associated with the aforementioned DCaC metadata identifiers (e.g., tags created and associated to different types of sensitive data as part of the DCaC process). Moreover, application **512** may include several microservices, some of which may contain a set of data sources **510** that may be utilized for training purposes.

(99) Additionally, the configuration may stipulate specific categories of data and/or data records in application **512** that should be used for the training. This input may determine the raw data that is required for the training (e.g., use only customer-related data, or focus on data related to orders, or focus on data related to a sensitive manufacturing process that is a trade secret, etc.).

(100) Further, the configuration may stipulate data sampling requirements, if any. For example, the configuration may stipulate how data should be selected, filtered, and/or sampled from data sources **510**.

(101) Furthermore, the configuration may stipulate an industry for which the model training will be performed. For instance, the configuration may specify that the model will be trained for the healthcare, finance, manufacturing, etc. industry. Additionally, the configuration may stipulate the geographical region where the data will be trained (i.e., the region where the data will be sent, processed, stored and potential retained). For example, the configuration may stipulate the geographical region of a centralized and compliant learning framework running in the cloud **528**. The configuration may also stipulate an identification of a cloud service to be used for the centralized and compliant learning framework running in the cloud **528** in the chosen zone/geographical region.

(102) Moreover, the configuration may stipulate a technique selected to generate compliant datasets before they can be sent to the centralized and compliant learning framework running in the cloud **528**. For instance, the configuration may specify that synthetic data creation, data anonymization, other possible techniques should be utilized, or that none of those techniques should be utilized,

etc. A different technique may be applied to each category of data depending on the characteristics of the datasets required for the training.

(103) At operation **544**, configurator **516** may utilize interface **530** to authenticate to and/or communicatively connect to compliance engine **306**. Configurator **516** may utilize the configuration elements stipulated in the configuration/request that it received from data training team **504** as inputs to compliance engine **306**.

(104) At operation **546**, compliance engine **306** may utilize these inputs to compute data compliance constraints to be applied to data collected at data collection node **506**. For example, based on these inputs, compliance engine **306** may use the identity of application **512** to access a corresponding data manifest bonded to the application **512**. Compliance engine **306** retrieve the categories of data handled by the application **512** (e.g., including indication of the metadata identifiers of each of the categories of data handled by the application **512**) by this access.

(105) Compliance engine **306** may also utilize the location of data collection node **506**, and use the industry, the geographical region, the identity of centralized and compliant learning framework running in the cloud **528**, the categories of data required for the training as provided as inputs from configurator **516**, etc. to compute the compliance posture and the corresponding data compliance constraints that apply in each case. Indeed, compliance engine **306** may compute the data compliance constraints considering the location of data collection node **506** where the application data is collected and/or the location of the centralized and compliant learning framework running in the cloud **528**.

(106) Compliance engine **306** may determine the capacity or lack thereof of compliant dataset generator **508** to store and use data from data source **510** depending on the computed compliance constraints. Even when compliant dataset generator **508** and application **512** are managed by the same organization (e.g., by two different units or teams within the same corporation or public entity), it may be critical to ensure that the purpose for collecting and using the data is compliant with data regulations.

(107) For instance, the primary purpose for using and storing the data from data source **510** in data collection node **506** may fall under the compliance limits of application **512**. However, the collection, storage, and processing of data by compliant dataset generator **508** might deviate from the primary (and compliant) purpose of using the data, and therefore, might require additional consents and/or the application of additional data compliance constraints with respect to those that already apply to application **512**. Hence, even when the management of the data, the models, and their corresponding training lie all under the same organization, the distinction between primary and secondary purposes in the usage and storage of data may be quite consequential from a compliance perspective.

(108) To determine whether the data can be used by compliant dataset generator **508** for model training purposes, compliance engine **306** may require explicit consent **502** to use the data for a secondary purpose. Consent **502** may be sought from and/or granted by the data team or application managers **304**, by a Chief Privacy Office (CPO), a Chief Compliance Office (CCO), a legal/compliance department, a user associated with the data, users who own the data or who the data is about, etc. Hence, compliance engine **306** may proceed to accept or reject the use of data from data source **510** for model training purposes.

(109) At operation **548**, compliance engine **306** may provide one or more training data constraints to compliant dataset generator **508**. The training data constraints may specify the list of metadata for the types of data required for the training, their corresponding associations to categories of sensitive data as well as a restriction flag indicating whether the data are legally restricted or not. In some examples where consent **502** is sought, compliance engine **306** may provide the training data constraints subject to ACCEPT==TRUE from the aforementioned consent determination.

(110) In various embodiments, the training data constraints may be provided to compliant dataset generator **508** as an array of tuples. The array of tuples may include the list of metadata for the

types of data required for the training, their corresponding associations to categories of sensitive data as well as a restriction flag indicating whether the data are legally restricted or not as determined by the data compliance constraints computed by compliance engine **306**. In some examples, the list of metadata in the tuples may include both predefined protected types (e.g., such as those specified in Table 1 and similar such types) as well as custom ones. For instance, an example array may be a list of tuples, such as: [(protected-type-1, Customer PII, RESTRICTED), (protected-type-41, Sales Confidential, UNRESTRICTED), etc.].

(111) Once processing is accepted and/or granted, configurator **516** may proceed with configuring compliant dataset generator **508**. At operation **550**, configurator **516** may configure data parsing and sampling module **514**. For example, configurator **516** may configure data parsing and sampling module **514** so that it is prepared to receive and handle the data coming from data sources **510**. This may include configuring the data parsing and sampling by data parsing and sampling module **514** according to the data sampling requirements specified in the configuration/request received from the data training team **504**.

(112) At operation **552**, configurator **516** may configure samples and metadata database **518**. Samples and metadata database **518** may be configured to store the production data requested for training the model.

(113) At operation **554**, configurator **516** may configure a training dataset generator **520**. Training dataset generator **520** may support core functions provided by compliant dataset generator **508**. Such a configuration may include the training data constraints defined in the array provided by compliance engine **306** at operation **548**. Once configured, training dataset generator **520** may resolve the categories of data (e.g., each identifiable by a metadata identifier) that it will need to process as well as their corresponding restrictions using these definitions. Once these configurations are completed, data collection from data sources **510** may begin.

(114) At operation **556**, the data collected from the various data sources **510** may be received by data parsing and sampling module **514**. At operation **558**, data parsing and sampling module **514** may select, filter and/or sample the data fed into compliant dataset generator **508** depending on the configuration of data parsing and sampling module **514** by configurator **516** at operation **550**.

(115) In various embodiment, compliant dataset generator **508** may expose an API to collect data from third-party applications, such as application **512**. The resulting data may be persisted, at operation **560**, for further processing in the samples and metadata database **518**. Some of the data stored in the samples and metadata database **518** may be stored in a non-compliant form (e.g., an original form, a non-anonymized form, a non-synthetic form, a form including sensitive data prohibited from direct use in model training, etc.). The different data segments may be indexed using the corresponding metadata identifiers as a key, i.e., by the annotations or tags in the application data as defined by the DCaC process described herein.

(116) Training dataset generator **520** may utilize the training data constraints, defined in the array provided by compliance engine **306** at operation **548** and configured to training dataset generator **520** by configurator **516**, to determine which data to process and how to process that data. For example, at operation **562**, each of the metadata identifiers might be used as a key to retrieve particular data segments from the samples and metadata database **518**.

(117) Training dataset generator **520** may, at operation **540**, determine whether a data segment (e.g., identifiable by a particular metadata identifier) is restricted data. For example, depending on a restriction flag (e.g., “RESTRICTED” or “UNRESTRICTED”) associated with a metadata identifier (e.g., as defined within a corresponding tuple) of a data segment, training dataset generator **520** may determine whether the data is classified in a restricted group, an unrestricted group, etc.

(118) When training dataset generator **520** determines that the data segment is classified in an unrestricted group, then the data may be persisted, at operation **564**, in an original (e.g., non-anonymized, non-synthetic, native, etc.) form in original non-restricted database **522**. The original

data may be stored in original non-restricted database **522** until it is sent to the centralized and compliant learning framework running in the cloud **528** for model training.

(119) When training dataset generator **520** determines that the data segment is classified in a restricted group, then the data may be sent, at operation **566**, to data indexing module **534** for further processing. The further processing may entail applying a variety of data processing and/or data transformation operations to generate compliant data sets from data that is not necessarily compliant for use in model training in its original format. For example, the further processing may include generating compliant data sets by selectively using synthetic data creation, data anonymization, or other techniques. Additionally, some of the data that is not compliant for use in model training in its original format may simply be excluded entirely from communication to the centralized and compliant learning framework running in the cloud **528** in any form.

(120) Restricted data segments may be indexed by data indexing module **534**. The indexed data may be correlated by data correlation module **536**. The correlation may ensure, for example, the synthetic data created mirrors the original data fields and patterns in the production data. For instance, all the occurrences of a specific customer name in the production data processed, may be systematically replaced by the same synthetic name. Similarly, all the occurrences of a specific customer address in the production data processed, may be systematically replaced by the same synthetic address, and so on.

(121) Compliant data generator **538** may then create compliant datasets for each category of data (e.g., identifiable by a corresponding metadata identifier). The compliant datasets may be created based on the configuration/request for training data received by compliant dataset generator **508** from data training team **504**. More specifically, the techniques specified in the configuration/request to generate compliant datasets before they can be sent to the centralized and compliant learning framework running in the cloud **528** may be used to select techniques such as synthetic data creation, data anonymization, or other possible techniques to be applied to each category of data.

(122) The resulting compliant datasets may be persisted at operation **568** in one or more databases. The database that the resulting compliant dataset is stored in may depend on the technique used to process the original data into the compliant dataset. For example, resulting anonymized data may be stored in an anonymized database **524** and/or resulting synthetic data may be stored in a synthetic database **526**.

(123) The compliant datasets may be sent, in **570**, to the centralized and compliant learning framework running in the cloud **528** for machine learning model training. This may include sending data from original non-restricted database **522**, anonymized database **524**, and/or synthetic database **526** to the centralized and compliant learning framework running in the cloud **528** for model training.

(124) As previously mentioned, the various elements illustrated in FIG. 5 may be distributed and/or run across different nodes. Also, the different levels of persistence described may be aggregated, simplified, or further segmented.

(125) In various embodiments, the compliant dataset generator **508** may be dynamically and/or automatically updated with any updates to data compliance regulations received by compliance engine **306**. For example, compliance engine **306** may send updates to configurator **516** (e.g., upon changes in data compliance regulations that may affect the constraints computed and/or the restriction flags).

(126) In addition, compliant dataset generator **508** may support multi-tenancy. That is, different applications and/or administrative domains may use compliant dataset generator **508** concurrently.

(127) In some embodiments, data sources **510** may include both data at rest and data in flight. Different techniques might be used to stream data to compliant dataset generator **508**. For instance, data may be streamed through a message broker, through a streaming API, etc.

(128) In various embodiments, the techniques described in relation to compliant data generator

538, synthetic database **526**, and/or anonymized database **524** for generating compliant datasets may be combined and further developed into a more granular processing. For instance, different techniques might be applied to individual data records within the same data segment (e.g., name.fwdarw.anonymize, weight.fwdarw.original, address.fwdarw.synthetic (within the same state), etc.).

(129) FIGS. **6A-6B** illustrate examples of model training deployments **600** for constraint-based training data generation according to various embodiments. The model training deployments **600** may be deployments of an architecture such as architecture **500** of FIG. **5**. Model training deployment **600** may include one or more data collection nodes **506** (e.g., **506-1 . . . 506-N**). Each data collection node **506** may be the computing node where an application **512** is executed. Application data from this execution may be collected from data sources **510** at the data collection node **506**.

(130) As previously outlined, the application data collected from the data sources **510** may be processed into and/or persisted locally to the data collection node **506-1** as compliant data. For example, the application data may be stored at a compliant data database **602** associated with the data collection node **506**.

(131) The compliant application data may be provided as training data to be used to train a machine learning model. For example, the compliant data may be sent to a centralized and compliant learning framework running in a cloud **528**. In various embodiments, such as is depicted in FIG. **6A**, the raw compliant data may be sent to the cloud **528** to train the machine learning model thereby leveraging the compute power and scalability offered by cloud infrastructures. Since processing by compliant dataset generator **508** result in only the compliant raw data being sent to the cloud **528**, data compliance concerns are ameliorated.

(132) In alternative embodiments, such as is depicted in FIG. **6B**, the compliant data used for the training stays where it is collected and possibly sampled, which may then be used by federated learning agents **604** (e.g., **604-1 . . . 604-N**) to train local models (e.g., at the edge). These local models may be combined into a global model at the cloud **528**, which may also leverage the benefits offered by cloud infrastructures. Since processing by compliant dataset generator **508** result in consent being automatically sought and/or obtained potential data compliance issues associated to the purpose and use of the data (e.g., secondary purpose issues) are ameliorated.

(133) FIG. **7** illustrates an example simplified procedure (e.g., a method) for constraint-based training data generation, in accordance with one or more embodiments described herein. For example, a non-generic, specifically configured device (e.g., device **200**), may perform procedure **700** by executing stored instructions (e.g., training data generation process **248**). In various embodiments, the device may include a data collection node. The data collection node may be a network node where the application is executed generating application data for model training.

(134) The procedure **700** may start at step **705**, and continues to step **710**, where, as described in greater detail above, a device may receive a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with metadata identifiers. The request may include an identifier of the application.

(135) At step **715**, as detailed above, the device may determine one or more training data constraints that restrict use of the application data as training data. The one or more training data constraints may be based on a data manifest bonded to the application that includes a listing of the metadata identifiers. The one or more training data constraints may be based on the identifier of the application. Further, the one or more training data constraints may be based on a location of the data collection node. Furthermore, the one or more training data constraints may be based on a location of a receiver of the training data.

(136) At step **720**, device may generate the training data in part by excluding application data of a particular type from being included in the training data based on a match between its metadata identifier and the one or more training data constraints. Excluding the application data of the

particular type from being included in the training data may include using the application data of the particular type to generate anonymized training data that is included in the training data. Additionally, excluding the application data of the particular type from being included in the training data may include using the application data of the particular type to generate synthetic training data that is included in the training data.

(137) At step **725**, as detailed above, provide the training data to be used to train a machine learning model. Additional steps may include obtaining, by the device, consent from one or more users for application data associated with those users to be included in the training data, and so on.

Procedure **700** then ends at step **730**.

(138) It should be noted that while certain steps within procedure **700** may be optional as described above, the steps shown in FIG. 7 are merely examples for illustration, and certain other steps may be included or excluded as desired. Further, while a particular order of the steps is shown, this ordering is merely illustrative, and any suitable arrangement of the steps may be utilized without departing from the scope of the embodiments herein.

(139) The techniques described herein, rather than dealing with data compliance as an afterthought, leverage DCaC metadata identifiers to automatically generate compliant application data sets for model training. The techniques described herein may be built into the application development lifecycle as well as in the data collection and sampling process. More specifically, this invention enables the collection and sampling of production (i.e., real) data handled by a DCaC-enabled application and automates the generation of compliant datasets that can be transferred to the cloud for centralized training and learning. This technique may be based on observing and sampling production data and their associated metadata (e.g., tags created and associated to different types of sensitive data as part of the DCaC process), along with a compliance-led dataset generation technique, which selectively combines synthetic, anonymized, and/or original data.

(140) While there have been shown and described illustrative embodiments that provide automated data compliance and observability, it is to be understood that various other adaptations and modifications may be made within the spirit and scope of the embodiments herein. For example, while certain embodiments are described herein with respect to using the techniques herein for certain purposes, the techniques herein may be applicable to any number of other use cases, as well.

(141) The foregoing description has been directed to specific embodiments. It will be apparent, however, that other variations and modifications may be made to the described embodiments, with the attainment of some or all of their advantages. For instance, it is expressly contemplated that the components and/or elements described herein can be implemented as software being stored on a tangible (non-transitory) computer-readable medium (e.g., disks/CDs/RAM/EEPROM/etc.) having program instructions executing on a computer, hardware, firmware, or a combination thereof. Accordingly, this description is to be taken only by way of example and not to otherwise limit the scope of the embodiments herein. Therefore, it is the object of the appended claims to cover all such variations and modifications as come within the true spirit and scope of the embodiments herein.

Claims

1. A method comprising: receiving, at a device, a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with a plurality of metadata identifiers; determining, by the device, one or more training data constraints that restrict use of the application data as training data; generating, by the device, the training data in part by excluding application data of a particular type from being included in the training data based on a match between a metadata identifier of the plurality of metadata identifiers and the one or more training data constraints; and providing, by the

device, the training data to be used to train a machine learning model.

2. The method as in claim 1, wherein the excluding the application data of the particular type from being included in the training data includes using the application data of the particular type to generate anonymized training data that is included in the training data.

3. The method as in claim 1, wherein the excluding the application data of the particular type from being included in the training data includes using the application data of the particular type to generate synthetic training data that is included in the training data.

4. The method as in claim 1, wherein the one or more training data constraints are based on a data manifest bonded to the application that includes a listing of the plurality of metadata identifiers.

5. The method as in claim 1, wherein the request includes an identifier of the application.

6. The method as in claim 5, wherein the one or more training data constraints are based on the identifier of the application.

7. The method as in claim 1, wherein the one or more training data constraints are based on a location of the data collection node.

8. The method as in claim 1, wherein the one or more training data constraints are based on a location of a receiver of the training data.

9. The method as in claim 1, further comprising: obtaining, by the device, consent from one or more users for application data associated with the one or more users to be included in the training data.

10. The method as in claim 1, wherein the device is the data collection node.

11. An apparatus, comprising: one or more network interfaces; a processor coupled to the one or more network interfaces and configured to execute one or more processes; and a memory configured to store a process that is executable by the processor, the process when executed configured to: receive a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with a plurality of metadata identifiers; determine one or more training data constraints that restrict use of the application data as training data; generate the training data in part by excluding application data of a particular type from being included in the training data based on a match between a metadata identifier of the plurality of metadata identifiers and the one or more training data constraints; and provide the training data to be used to train a machine learning model.

12. The apparatus as in claim 11, wherein the application data of the particular type is excluded from being included in the training data by using the application data of the particular type to generate anonymized training data that is included in the training data.

13. The apparatus as in claim 11, wherein the application data of the particular type is excluded from being included in the training data by using the application data of the particular type to generate synthetic training data that is included in the training data.

14. The apparatus as in claim 11, wherein the one or more training data constraints are based on a data manifest bonded to the application that includes a listing of the plurality of metadata identifiers.

15. The apparatus as in claim 11, wherein the request includes an identifier of the application.

16. The apparatus as in claim 15, wherein the one or more training data constraints are based on the identifier of the application.

17. The apparatus as in claim 11, wherein the one or more training data constraints are based on a location of the data collection node.

18. The apparatus as in claim 11, wherein the one or more training data constraints are based on a location of a receiver of the training data.

19. The apparatus as in claim 11, wherein the process when executed is further configured to: consent from one or more users for application data associated with the one or more users to be included in the training data.

20. A tangible, non-transitory, computer-readable medium storing program instructions that cause a

device to execute a process comprising: receiving, at the device, a request for training data that is based on application data generated by an application executed at a data collection node, wherein the application data is associated with a plurality of metadata identifiers; determining, by the device, one or more training data constraints that restrict use of the application data as training data; generating, by the device, the training data in part by excluding application data of a particular type from being included in the training data based on a match between a metadata identifier of the plurality of metadata identifiers and the one or more training data constraints; and providing, by the device, the training data to be used to train a machine learning model.
